IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART I: MEng, BEng and ACGI

**Corrected Copy**

**SEMICONDUCTOR DEVICES**

Thursday, 6 June 10:00 am

Time allowed: 2:00 hours

**There are THREE questions on this paper.**

**Answer ALL questions.**

*Question One carries 40% of the marks. Questions Two and Three each carry 30%.*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : K. Fobelets
Second Marker(s) : S.W. Wright

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**Constants**

| | |
|---|---|
| Permittivity of free space: | $\varepsilon_o = 8.85 \times 10^{-12}$ F/m |
| Permeability of free space: | $\mu_o = 4\pi \times 10^{-7}$ H/m |
| Intrinsic carrier concentration in Si: | $n_i = 1.45 \times 10^{10}$ cm$^{-3}$ at $T = 300$ K |
| Dielectric constant of Si: | $\varepsilon_{Si} = 11$ |
| Dielectric constant of $SiO_2$: | $\varepsilon_{ox} = 4$ |
| Thermal voltage: | $V_T = kT/e = 0.026$ V at $T = 300$ K |
| Charge of an electron: | $e = 1.6 \times 10^{-19}$ C |
| Planck's constant: | $h = 6.63 \times 10^{-34}$ Js |
| Bandgap of Si: | $E_G = 1.1$ eV at $T = 300$ K |
| Effective density of states of Si: | $N_C = 3.2 \times 10^{19}$ cm$^{-3}$ at $T = 300$ K<br>$N_V = 1.8 \times 10^{19}$ cm$^{-3}$ at $T = 300$ K |


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**Formulae**

$$f(E) = \frac{1}{1+\exp\left(\frac{E-E_F}{kT}\right)}$$ Fermi distribution

$$n_i = \sqrt{N_V N_C}\exp\left(\frac{-E_G}{2kT}\right)$$ Intrinsic carrier concentration

$$n = N_c \exp\left[\frac{(E_F - E_c)}{kT}\right]$$ Concentration of electrons

$$p = N_v \exp\left[\frac{(E_v - E_F)}{kT}\right]$$ Concentration of holes

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon}$$ Poisson equation in 1 dimension

$$\left.\begin{aligned} J_n(x) &= e\mu_n n(x)E(x) + eD_n\frac{dn(x)}{dx} \\ J_p(x) &= e\mu_p p(x)E(x) - eD_p\frac{dp(x)}{dx} \end{aligned}\right\}$$ Drift and diffusion current densities in a semiconductor

$$I_{DS} = \frac{\mu C_{ox} W}{L}\left(\left(V_{GS} - V_{th}\right)V_{DS} - \frac{V_{DS}^2}{2}\right)$$ Current in a MOSFET

$$\left.\begin{aligned} J_n &= \frac{eD_n n_{p_0}}{L_n}\left(e^{\frac{eV}{kT}} - 1\right) \\ J_p &= \frac{eD_p p_{n_0}}{L_p}\left(e^{\frac{eV}{kT}} - 1\right) \end{aligned}\right\}$$ Current densities for a pn-junction with lengths $L_n$ & $L_p$

$$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_A N_D}{n_i^2}\right)$$ Built-in voltage

$$c = c_0 \exp\left(\frac{eV}{kT}\right) \text{ with } \begin{cases} c = p_n \text{ or } n_p \\ c_0 \text{ bulk minority carrier concentration} \end{cases}$$ Minority carrier injection under bias $V$

$$D = \frac{kT}{e}\mu$$ Einstein relation

$$w_n(V) = \left[\frac{2\varepsilon(V_{bi} - V)N_A}{e(N_A + N_D)N_D}\right]^{1/2} \& \; w_p(V) = \left[\frac{2\varepsilon(V_{bi} - V)N_D}{e(N_A + N_D)N_A}\right]^{1/2}$$ Depletion widths under bias $V$

$$W_{depl}^{\max} = 2\left[\frac{\varepsilon kT \ln\left(\frac{N_{substrate}}{n_i}\right)}{e^2 N_{substrate}}\right]^{1/2}$$ Maximum depletion width


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# The Questions

1. Basic principles of semiconductors

a) Different physical parameters influence the conductivity of a material.

i) Name the two parameters that influence the electrical conductivity of a material. [2]

ii) How does the conductivity of an intrinsic semiconductor change when increasing the temperature from room temperature (300 K) to 400 K? [2]

iii) Give the reason for your answer in ii). [4]

iv) How does the conductivity of a metal change when increasing the temperature from room temperature (300 K) to 400 K? [2]

v) Give the reason for your answer in iv). [4]

b) Different carrier transport processes can determine the currents.

i) Name the carrier transport processes in a Cu conductor carrying a current. [2]

ii) Name the carrier transport processes that can occur in a forward biased Si pn diode. Identify the carriers and processes in each region. [6]

c) Energy band diagrams.

i) Draw the energy band diagram with $E_c$ & $E_v$ and add $E_F$ to represent a homogeneously doped p-type Si layer under a bias of 1V (ground node at left and +1V at right). Indicate the voltage difference on the plot. [4]

ii) Give the value of the potential energy difference for the holes across the p-type Si layer in i). [2]

iii) Give the value of the potential energy difference for the electrons across the p-type Si layer in i). [2]

d) Capacitor 1 is built with a metal top and bottom plate and a 100 nm $SiO_2$ insulator layer. Capacitor 2 has the same metal top plate, a 100 nm $SiO_2$ layer and an n-Si bottom plate with a doping density that generates flat band at zero bias. The plates of the capacitors are parallel to each other and are 500 μm thick and their surface area is 100 $\mu m^2$. Both capacitors are characterised for DC voltages in a range of $-2\ V < V_{DC} < +2\ V$ and with a small 20 mV ac signal of 1 MHz.

i) Sketch the capacitance-voltage characteristic of capacitor 1 for $-2\ V < V_{DC} < +2\ V$. Give the value of the maximum capacitance. [5]

ii) Sketch the capacitance-voltage characteristic of capacitor 2 for $-2\ V < V_{DC} < +2\ V$. Give the value of the maximum capacitance. [5]


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2. MOSFETs

a) Give the definition for the threshold voltage of a MOSFET. Ensure your definition is applicable to both n and pMOS. [4]

b) The energy band diagram from gate through the oxide into the silicon (MOS).

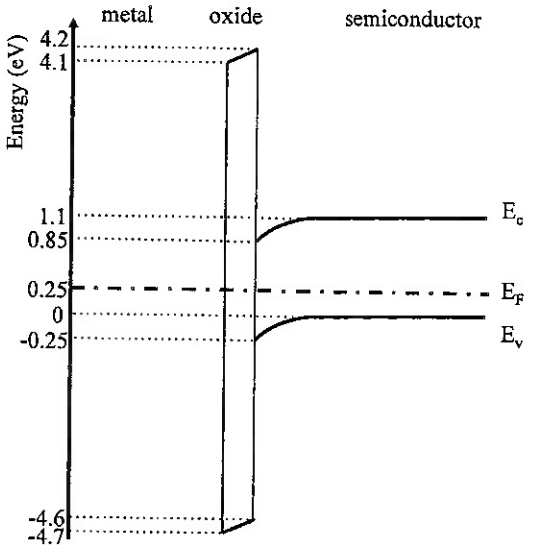



Figure 2.1: the energy band diagram of an MOS structure.

i) Give the doping type in the silicon. Explain your answer in 1 sentence. [2]

ii) What is the gate voltage on the metal contact? Explain your answer in 1 sentence. [2]

iii) Extract the workfunction difference between the metal and the silicon from figure 2.1. [4]

iv) Is the workfunction of the metal larger, smaller or equal to the workfunction of silicon? Explain your answer briefly. [4]

v) How would the threshold voltage change if the bending of the energy bands in all layers in figure 2.1 remain the same by adapting the material choice for the metal and decreasing the doping density in the semiconductor? Explain your answer briefly. [4]

c) For the MOS in figure 2.1, sketch the capacitance-voltage characteristic ($C_{GS}$ as a function of $V_{GS}$. $V_{DS} = 0V$). Identify the accumulation, depletion and inversion region. Give the value of the voltage where the capacitor goes from accumulation into depletion. Identify the voltage point with the symbol $V_{th}$ when the capacitor goes into inversion. [10]


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3. BJT

Table 3.1 pnp BJT parameters.

| | E | B | C |
|---|---|---|---|
| Doping type | $N_A$ | $N_D$ | $N_A$ |
| Doping density ($cm^{-3}$) | $10^{18}$ | $10^{16}$ | $10^{16}$ |
| Layer width ($\mu m$) | 1 | 0.8 | 2 |
| Minority carrier diffusion length ($\mu m$) | 5 | 20 | 50 |
| Minority carrier diffusion constant ($cm^2/Vs$) | 20 | 12 | 36 |

a) Draw the energy band diagram ($E_c$, $E_F$, $E_v$) from emitter through base into the collector of the pnp bipolar junction transistor in forward active mode (FAM). Ensure the relative magnitude of different parameters in your plot is correct. [10]

b) Give the full expression for the diffusion current densities in the pnp transistor without neglecting components. Then calculate the currents, identifying and making appropriate assumptions to simplify your calculations (+ ignore depletion widths). The bias conditions are $|V_{EB}| = 0.5V$ and $|V_{BC}| = 1V$.

i) the emitter current density. [4]

ii) the collector current density. [4]

iii) the base current density (apply simplifications directly). [2]

iv) the current gain (apply simplifications directly). [2]

c) Output characteristics.

i) Sketch the output characteristics of a pnp BJT in common emitter connection for three different and equally spaced values of the base current. Take the impact of the depletion regions across the junctions into account. [3]

ii) Draw the energy band diagram of a BJT in FAM at a high value of $|V_{BC}|$ that depletes the base region completely. [3]

iii) Referring to the answer in ii) what will happen to the output current? [2]


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